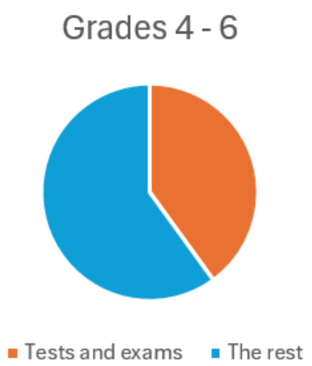
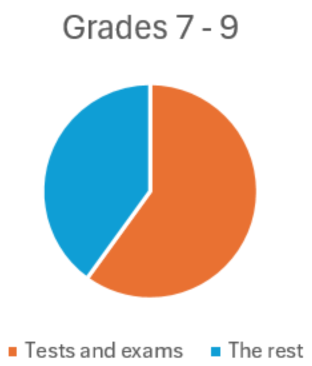
Study Smarter: Guidance for completing the Student Workbook

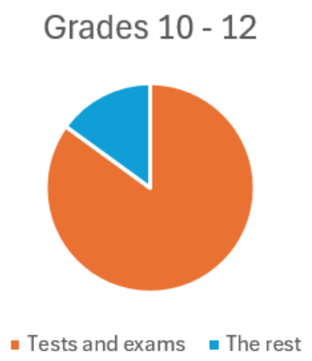
In this guide I'll give you the answers to the questions in the Student Workbook, advice for how to coach the student, and links to some of my go-to resources. The book is a starting point for you which is why I've perforated the pages. When you tackle a page tear it out, paste it into your Study Smarter notebook and you have all the space you need to answer or add further resources. The Workbook and your notebook clearly show what you have done so it's easy to do part of the book in one year and continue in the next. I would however recommend sticking to the order of the book as everything builds on the previous. You can even digress and expand on certain areas. For example, you could digress into the types of memory and our working memory capacity. You're the coach; enjoy it!

PART A Getting Motivated

P7







In the tick exercise, most of not all of the items will likely be ticked.

P9

1.1 Most teens have a rather fixed mindset when it comes to academics. But there are great stories in SA each year of students who have exhibited huge self-belief and effort.
1.2 Yat Pan had a growth mindset
1.3 Scales like this (especially 5-point scales) are very common in research. Traditionally they have a neutral midpoint.
1.4 This is so hard   Must we do this? I don't know what to do.   I can't do Maths.   I'm going to get an exemption for Afrikaans   This is so boring   My teacher hates me
All of these point to finding the easy way out, or an excuse, when things seem hard.

P11

In this section we encounter the first of many questions that require students to reflect, or think about their thinking, which is called metacognition. Schools do not often teach the skill of reflection and the frontal lobe of the teen brain which deals in part with metacognition (and executive function – planning, anticipating, foreseeing, organising) is far from mature. Metacognition is therefore likely to be a new think and must be seen as a skill all on its own to be taught and developed.

Its not mentioned in this section but this a good time to look at bad vs good habits and what makes them so. Consider bad habits as those that take little or no control.

2.1 They …

- made primitive tools. These allowed them to do more things that they couldn't do with bare hands – cut, crush, skin, etc.
- controlled animals by domesticating them. We know modern humans outcompeted Neanderthals. Scientist think this was the result of man domesticating dogs , which allowed them to hunt more effectively and protect themselves.
- They took control of their need for food. They moved with migrating animals, learnt how to harvest and forage for food and what could be eaten vs not.
- They took control of health and hygiene. They knew what substances had medicinal powers.

2.2 Open ended. Teens like to be in control of their choices and actions – what they do, when they do it and how. Of who their friends are, the clothes they wear, their tech choices, what time they get up, how long they spend on screens. Maybe subject choice going into Grade 10. If they could, who their teachers would be and how they would structure school.

2.3 Many teens struggle with identity issues. Maybe they feel trapped by parental or school rules. Being overwhelmed in another way of feeling out of control. So things not going the way they want. Schoolwork going badly. Constantly getting in trouble due to bad habits.

2.4 Most teens would like all the trappings of success or power. More realistic though, is getting the results they ought to get for the effort they put in. Try to direct conversation to things that are realistic.

P13

2.4 In my control? No … BE POSTIVE (in biiiig letters!) Yes… TAKE CONTROL

Note: It's worthwhile unpacking what being positive looks and sounds like, versus what being negative. Teens can be remarkably negative. It's normal and a survival instinct that resists change, causes slow adoption, is cautious, and is pessimistic especially when one is unsure of oneself. In such situations negativity can maintain the status quo and keep us from doing things we are unsure of.

2.5 Re tests :

In my control	Out of my control
How much I study What I study The stationery/equipment I bring into exams My study plan My health – diet, exercise, sleep How much I listen in class to the test content How much I ask questions about the work How much I practice for the test How much I catch up missed work etc	Who set the test What's in the test When the test is. Is it a busy time? How long it is (time and marks) How the questions are phrased Unexpected things that interfere with studying etc

P15

3.1 Examples

- I was late for school because of the traffic
- I didn't do it because I didn't hear you/I was talking to Joe
- I didn't know what to do (in the context of HW not done)
- I couldn't do them (in the same context)
- I didn't know I had to do it (in the context of anything not done)
- They're in the wash/broken (in the context of incorrect uniform items)
- Everyone was doing it (in the context of some misdemeanour)
- Etc

3.2 In almost all cases the student could have done something about it they really cared to. For example, being late for school might be traffic BUT the family might only get caught in traffic because the student is tardy in the morning. Being able to have an effect or control is have agency, or 'to be agentic'. The tendency to opt for an excuse is because it is usually the easier option and removes consequences.

3.3 In the short term it might seem clever. In the long term, it enables unconstructive behaviour, erodes agency, and along with it, self-esteem. Others also build a poor view of you; it erodes their trust in you, and you are seen as undependable. A widespread feeling of lack of control is a feature of depression. Giving up control must not be taken lightly if we consider mental health.

3.4 Who's responsibility is it to wake up on time? When you are young your parents take responsibility. As you get older it's a sign of growing up that you take more responsibility for yourself. Set your own alarm. If your phone is not allowed in your room get your parents to buy you a clock. Are you over tired? Are you staying up too late or not exercising enough? These are all things you can so something about and are not valid excuses. Generally the only valid excuses apply to exceptions; when life simply gets in the way of good plans and intentions.

P17

4.1 You might be satisfied with the outcome of a dispute (over homework not done, or a class incident), with how something you were making has turned out, after eating a generous meal, how you did in an event, how a plan worked out.

4.2 Fulfilment is a deep, inner satisfaction. Fulfilment is rarely felt, so expect answers of last week, last month, last year, I can't remember. Circumstances: Fulfilment generally comes when effort has matched challenge. It's often limited to musicians, artists and sportspeople, those who got through a tough task, or people who master a hectic schedule.

4.3

	Example	Words
Ex0	Studying but getting a disappointing mark … again Repeatedly failing at a game level Not making the grade for something I applied for Missing out on my university first choice	Disappointed Frustrated Angry Sick and tired Burnt out So over it I'm hopeless
0XA	Hopelessly outclassing another school in a team sport Having to do tasks or challenges way below your age range	Bored Disinterested Whatever It's a joke I behaved unkindly
ExA	Getting the mark you wanted in a test after hard studying Feeling like you really improved/learnt something in a tough sports practice At last gasping a difficult concept	Hoo-haa! I'm unstoppable I'm the BOSS! YES!!! That really meant something to me That felt really good

4.4 This is almost certainly the Ex0 block for all but the rare exception. Even top achievers are often only 'pleased' with results as opposed to really fulfilled. This is because they often put in HUGE effort; they get the results but it's sheer slog and not pretty. Lots of distinctions sounds great but distinctions are by no means rare where education is sound so it's maybe not *that* great.

4.5 Daily is key. School lends itself to fulfilment because it is filled with many milestones to achieve each day. Because fulfilment is so rare but so powerful every opportunity must be sought get regular, small, easy injections.

Examples: I would feel really good I could …

- Get through a day with all my homework done
- Understand everything I hear in class
- Avoid getting any demerits (where I often do)
- Remember all my books
- Answer what the teacher asks in class
- Score a goal in breaktime soccer

P21

5.1 Procrastination often offers short term relief (but longer term problems). Psychologists say that a lot of human behaviour is explained by trying to avoid pain. Procrastination avoids immediate pain. The key is to make the pain of delaying things more painful that doing what needs to be done now.

Examples:

- Putting off homework until later
- Putting off studying until tomorrow
- Starting a project tomorrow

The reward is usually immediate relief and avoiding thinking about what worries us. In all of these the relief will feel good. But how did you feel later when the consequences caught up?

5.2

- I was scared of going to school camp but I made myself and it felt great.
- I didn't want to enter for a sports day event but I did and I'm proud of myself.
- I was scared of speaking to my teacher but I did and it went far better than expected
- I was scared of speaking up about what I saw but I did and things turned out well
- I didn't feel like studying but I made myself and it felt good afterwards
- My room was a mess. I tidied it and it feels so much better.

5.3 The tests refer to things that the student isn't doing, or isn't doing well, but could be. Start small, with things that might matter.

- I don't like asking questions in class. I could make myself ask at least one question in each lesson.
- I struggle with homework. I could try do it all for just one week and see how it feels.
- I try and get work out of the way as fast as possible. I could go over all my Maths answers for just this week.

5.4 Be prepared to discuss what comes up. See 5.1 for the main rationale. Procrastination is essentially avoidance. But avoidance strategies create psychological pain, which risks anxiety and depression. Remember the effect of not taking control when we have control.

Examples:

https://www.counseling.org/publications/counseling-today-magazine/article-archive/article/legacy/procrastination-an-emotional-struggle

https://www.psychologicalscience.org/observer/why-wait-the-science-behind-procrastination

P23

1. 1.1 F 1.2 T 1.3 F 1.4 T 1.5 F 1.6 T

2. 2.1 take control 2.2 fulfilment 2.3 effort x achievement 2.4 ownership 2.5 follow

3. This is the first reflection task. It might be worthwhile pausing here to unpack reflection a bit further. The questions each have prompts to help do this. Group feedback on this will be helpful. Honest answers are important here. Respect honesty and keep the student feeling 'safe; while giving direction where needed. It is better to ask questions more than direct or give answers. When people arrive at answers for themselves they are more likely to be receptive to them.

PART B Goals and action plans

P25

1. The goal will almost certainly be along the lines of "I want to get X in subject X", or more vaguely, I" want to do better in subject X". Others might be "To make Team X in sport X", "To make WP champs in X". Whatever it is it will almost certainly be phrased as a rather distant 'destination' – where I want to be.
The APs will be new to many as goals, once set, tend to be fuelled by hope alone, rather than

action. If the goal is to get 70% for Maths, what I do today obviously has to do with why I'm not getting 70% already; what I'm doing wrong at the micro level. This automatically calls for personal interrogation and leads to the identification of key, actionable problems and their possible solutions.
Example: I think I'm not getting because … I don't always copy the notes off the board, I don't ask questions when I don't understand, I spend too much time chatting to my friend in class, I give up too easily on my homework, I don't do enough practice examples to solidify my understanding, etc.

P27

2.1 It is:

- S: This 100-piece puzzle
- M: It's achieved when I complete the puzzle. I know when its complete.
- A: I reckon a 100-piece puzzle is well withing my reach. In fact, it may be a little too achievable
- R: Relevant: The task seems to have a purpose so doing the puzzle is relevant. NB: Relevant often is the reflection point. Remember the '5 whys' – *Why does this matter?* To get to the heart of issues.
- T: Yes. I have 15 mins
- A: It wasn't stated but it's likely I'll be accountable to the teacher for how far I get.

2.2 Possible answers below. Note these all happen before the goal is reached and help achieve it step by step:

- Start by turning all the pieces face up.
- Complete the outside pieces first.
- Keep referring to the box.
- Work on distinctive areas, such as a particular colour or pattern

2.3 They probably weren't SMARTA. Where a mark is set this is measurable but that falls away with relative language such as 'do better'.

2.4 It's worthwhile spending some time on this as students won't be used to SMARTA or action plans. It takes some time to get it right.

The Maths goal and APs (identified as the relevant issues) might now be:

Goal: *To get 70% (where I was getting 65%) in both Maths tests this term*.

APs:

1. To copy work fully off the board every time worked examples are given, even if it means taking a pic and completing at home. Show mom and dad each time.

2. To ask questions/for help whenever I don't understand. If it happens in HW I will at least attempt the questions but identify the sticking point to ask my teacher rather than guessing. I'll let my teacher know I'm doing this.

3. Do at least 30 min practice on Maths app each day of the current topic. Tell mom and dad when I'm done daily.

PART C All about learning

P29

1.1 Identifying this is again a matter of reflecting accurately. It might be that I don't learn in class effectively; I don't do my HW properly and I certainly don't go over work on a daily basis if it's not assigned; I struggle with studying – it doesn't seem to give the desired results; I don't know how to make proper study notes (though I make lots); I don't do enough past papers and when I do I'm not sure if I'm doing them properly; I struggle in the actual assessment task; I usually get my mark back but don't think over the link between my mark and my behaviour. Note that students will find it difficult to identify problems; it's hard to know what you don't know if you're not use to it. Learning to meet own needs is to be self-regulated and resourceful.

1.2 This is one of so many examples you can use. In it you should see the two women. Have some fun a see if you can google other examples. Search for "ambiguous pictures" (involving images) or "ambigrams" (involving words). Note how hard it can be to see them at first, but how hard it is not to see them once you have! You can also look for optical illusions. The takeaway here is that once something is 'learned' – once you 'get it' – it is not easily forgotten.

1.3 This involves considering yourself from the perspective of an outsider, one who just sees actions without all the 'reasons' that we have in our own head. Be honest with yourself. For example: My teacher might see me as someone who talks too much in every lesson and always has something to sat about everything. I can't mind my own business and focus. I think I need to think about having more self-control. It'll help me, and if I tell the teacher I'm working on it, maybe together we'll develop like a coaching relationship and my teacher will respect me more.

Some parents do not allow their child to see their Report comments. I think they should see them even if they're a blow. Facing up to reality is healthy and part of taking ownership. It promotes discussion around control again and goals etc. Bear in mind the report comments are often very tactful; the hard truths are often between the lines.

P31

I illustrate attention and distraction as per my book. I get students familiar with a simple song (9 green bottles, 5 little monkeys, etc). They have to sing the song while reading some projected text. The chaos illustrates how you can't do two things involving the same brain area – word processing in this case - at once. Many students are distracting/distracted in class. Talk about that. I also use the example from the book about the moonwalking bear awareness test. See https://www.youtube.com/watch?v=xNSgmm9FX2s . If you're distracted from the important you will miss it. The brain is smart, but attention means we only attend to a part of what we sense.

2.1 A good reflection exercise. Watch for excuses or blame shifting here. Make the student see *their* role in solving whatever the issue is. Get them to imagine themselves in the lesson they had *today*.

2.2 Very likely yes. Very likely there is a teacher who 'hates me'. This needs to be unpacked. How do you know they hate you? What might I be doing that makes them react to me as they do? How must it feel dealing with this as the teacher day in and day out? Can I blame the teacher for how they respond to me? Students must be encouraged to see teachers as coaches (hopefully your child is fortunate enough to have teachers who care). If there's an issue, take control. Request a meeting with that teacher and play open cards where you're at. Hear them out and work out a plan to move forward. Or ask another teacher you trust for help/advice. If the teacher issues remain, then consider that the worse the teacher is the better you have to be, so be positive and be agentic. Don't let the teacher win. Solve your own problems.

2.3 'I' language is disarming; 'you' language is accusatory and causes defensiveness. The examples below will be VERY unusual … but they are the kind if exchanges that should be happening in safe environments where students are serious about their studies and their relationships. Examples:

- I really struggle to follow your reasoning sometimes. I'd find it really helpful if you could explain to me in some more ways.
- I felt that the demerit today was unfair, and I was hoping we could chat it through. I felt I wasn't doing anything wrong. I just wanted to understand what got me the demerit.
- I don't want to be rude or anything, but I feel you hate me. I was hoping we could chat about because I really want to do well in your subject.

Class skills (the third think to think about in learning) refers to organisation, planning, comprehension, subject-specific skills, critical thinking, interpersonal skills, etc, etc. Important and worthwhile thinking about if they're the issue but beyond the scope of this book.

2.5 This exercise is yet another attempt to get the student to take an outsider perspective on themself. Seeing that others 'have a point' is a big step towards a more understanding, forgiving and constructive attitude. Q4 might be affected by how much I like the subject, how much fulfilment I get from the subject, who is in the class with me, the physical environment, and how the teacher manages the class. Which again leads to do I want to do well in the subject? So, what is in my control and what am I going to do about it? Is there an AP here somewhere?

P37

3.1 Lateral answers accepted! I would be able to:

- Think insanely fast.
- Remember everything that I ever saw, heard, or smelt.
- Read other people's minds.
- Focus on two things at once.
- Understand everything instantly.

3.2 It's extremely rare to talk about something without some images coming to mind. Often, we aren't even conscious of the images. This task draws attention to our thinking and the imagery we so often take for granted. More images will come to mind if we are asked detailed questions about our story. The 1-minute timing encourages focus. Make thing of it.

P39

There are many examples to show. Google "pareidolia clouds" and you'll get lots. Better yet, Google "pareidolia plants" and you'll get more real examples. The odd thing here is if you ask students what do you see, they are least likely to say ‘clouds’ or ‘plants’. The brain takes the obvious for granted and rather focusses on the cryptic, the hidden, the interesting. There are some nice videos to illustrate natural curiosity in animals. See this great example https://www.youtube.com/watch?v=juWNZGSW-_M

3.3 There are lots of examples:

- Tourists visiting the sites
- In computer games we want to know what’s on the next level, or in the next room
- Many films keep us hooked by making us wonder what will happen next.
- We like twists and turns – think detective/crime movies
- It’s why we do puzzles, solve crosswords and riddles
- It’s what we when we go somewhere new; we take it all in
- It’s what makes internet clickbait so attractive (check out some examples/ For example, “The doctor took one look at her and called emergency. What happened next was shocking”)

3.4 This is my take on the psychologists’ tool designed to elicit our most basic and instinctive response, which may be driven by our most deeply ingrained thought patterns. The exercise offers insight into a person’s core ways of thinking and existing knowledge patterns. The game is an introduction to an important aspect to studying: that we need to leverage our natural associations and incorporate new information into existing patterns rather than keeping it isolated

P41

3.5 a) this is 12 ‘bytes; as individual items. But seeing it as 5 repeats of 12 shortens it to 3 bytes

b) saying 10 numbers in a strict monotone makes them hard to remember. This is why we intone numbers in groups: 084 322 4333. This intonation is easier to remember.

3.6 This example illustrates how I use how *my* mind works to help *me* remember:

- Woodhead is where I started. By drawing a head like a log.
- Then Victoria. I made the head belong to my image of Queen Victoria – a stern, unhappy image.
- De Villiers. Royalty holds a sceptre which I reimagined as a devil’s fork… which triggers the name
- Hely Hutchinson. Royalty is ‘Her Royal Highness’ … which reminds me of HH and the name
- Albert. Albert was the insignificant other who here is to the side of the queen.

The three things we remember:

Significant: See this example: https://www.youtube.com/watch?v=lpL0Th49qVQ The monkey exhibits curiosity with this new object that turns out to be a mirror. The reflection catches the monkey way off guard. Its likely not to forget this event/object, just we don’t forget significant things in our lives. What makes things significant at school? Anything that stands out as distinctive.

Interesting: We all know people who have an interest in something and in that case how much they know about it. Think soccer fans and how much detail they remember.

Repeated: We all know how repetition helps us remember. It is the one and almost only tool that most study methods rely upon.

3.7 These are personal. Examples:

Significant: *I can picture the notes on X I made in red in my notebook*

Interesting: *I like X and I know a lot about it*

Repeated: Repetition works whether we think about it or not. We all know the lyrics to many songs without even trying to commit them to memory. It just happens. We remember the route to places, or where shops are in the mall. We recognise company logos we've seen often.

P43

1. So that the participants couldn't use existing associations to help them remember the words. Nonsense words have no links or reminders to help remember them. His results were based purely on rote memory
2. Words were forgotten after they had been memorised (All the graphs slope downwards to the right). In other words, all memory erodes over time
3.

	Event 1	Event 2	Event 3	Event 4
% remembered	80	87	91	95

4. Each times words are memorised, the memory lasts longer, or is better after the same time interval
5. a) zero. The graph shows they would have forgotten all the words by Day 3
   b) Thanks to their three revisions, they know 95%
6. Going over work before you forget too much of it makes you remember it for longer. Not going over it means you are likely to have forgotten pretty much all of it by the time a test comes. This means you have to do a LOT of work to memorise it, and you are likely to cram. This means you should go over your work daily as a start. Important: The research says that in going over work you should try to <u>recall</u> it before you read it.
   Don't over the work lots of times in the same session; it's more powerful to do it once for a number of sessions. This is spaced repetition and there is lots of evidence for the effect.

7. Event 1: Paying close attention to work in class where initial learning takes place
   Event 2: Going over the work the same day after school
   Event 3: Going over the work on the first weekend
   Event 4: Going over the work as part of your revision schedule, perhaps just before you start studying intensively for the test

P47

1.
   - 1.1 T (true unless you are very self-reliant)
   - 1.2 T (both are words based)
   - 1.3 T (it depends on all the things in the learning diagram)
   - 1.4 T (think back to the picture illusions)

- 1.5 T (actually a big, and often the deciding, part)
- 1.6 we covered two groups of three
- 1.7 T
- 1.8 F There is no evidence that more complicated, bigger pictures are harder to remember. This is partly because the more complex the picture the more associations we build PROVIDED we are an engaged.
- 1.9 T Shown by Ebbinghaus and backed up by everyday experience
- 1.10 T BUT before reading it you must first try recalling it.

2.

- 2.1 modern world
- 2.2 survive
- 2.3 pictures
- 2.4 curious
- 2.5 associations
- 2.6 chunking
- 2.7 significant
- 2.8 interest
- 2.9 repeated

PART D Study strategies and tools

P49

1.1 Yes, except for summarise. Imagine you were reading text in a foreign language. You can highlight any bold words, read it (clumsily perhaps), recite it in short bits as you read it, write it out, and memorise it. To summarise it (shorten it) you would need to at least comprehend it … unless you just left out random words!

1.2 It's natural to struggle to get down to studying. If the work doesn't mean anything to you and you study passively you will get bored. Boredom comes with distraction as the brain searches for something more interesting or relevant.

Boys tend to be less diligent and go for quick fixes. They tend to do things more verbally. Girls are usually more diligent. They are more likely to write out notes (often neater than the original!) and pend time making things pretty.

P51

2.1.1 Not really. Acronyms work best when you have flexibility to play around with letters. The planets would be MVEMJSUN which looks and sounds meaningless. No association comes to mind. It's not impossible though: You could conjure up the image of wise old Indian astronomer called Mvemjay-san. See 2.1.3

2.1.2 You could use any order of letters as the order doesn't matter. ERIC might mean something. The common option is RICE.

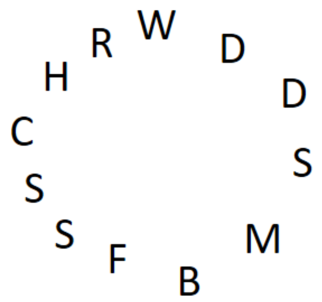
2.1.3 The colours of the rainbow, which do have an order: Red Orange Yellow Green Blue Indigo Violet

2.1.4 Radio detecting and ranging. Note this acronym does not stick to rule of first letters. And it even uses 'A' from 'and'. See also SCUBA, NATO, LASER, etc, etc

P53

2.2.1 There are 12 characteristics. Arguably too long for an acronym. The order doesn't matter so we have flexibility. First, I bolded what I regarded as my key words. Then I arranged the first letters in a circle as on the next page. Then I got creative and played with letters to arrive at the final acrostic.

- They **worked** slowly
- They **disobeyed** their masters
- They got **drunk**
- They committed **suicide**
- They **murdered** their owners
- They **broke** tools
- They set **fire** to crops and houses
- They pretended to be **sick**
- They **stole** food from the masters' kitchens
- They were **cheeky** (disrespectful) and answered back
- They **hurt** themselves on purpose
- They **ran away**



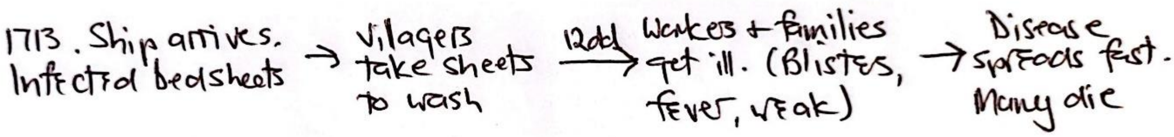
My acrostic: *How Badly We Make Roast Corn. First SSSit DDown*!

For: *Hurt | Broke | Worked slowly |Murdered | Ran away | Cheeky |Fired | Suicide |Stole |Sickness |Drunk I Disobeyed*

Note that the acrostic is only vaguely sensical. It brings to mind slaves sitting around a fire roasting corn cobs. But they have to sit down first. And it's made by a stammerer. Anyway, it works!

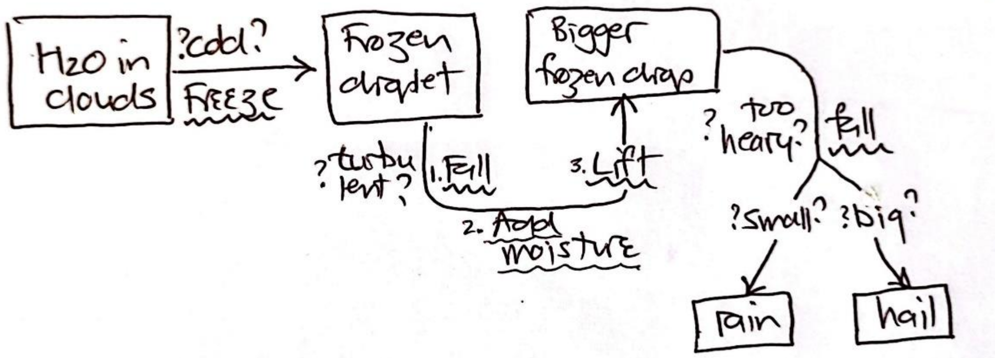
P57

2.5.1 Here is a possible diagram. It's a linear sequence of event so I've used a linear format. There is no need to use blocks. Note also everything is free hand – no rulers. Using a ruler is a distraction from engaging with the content.



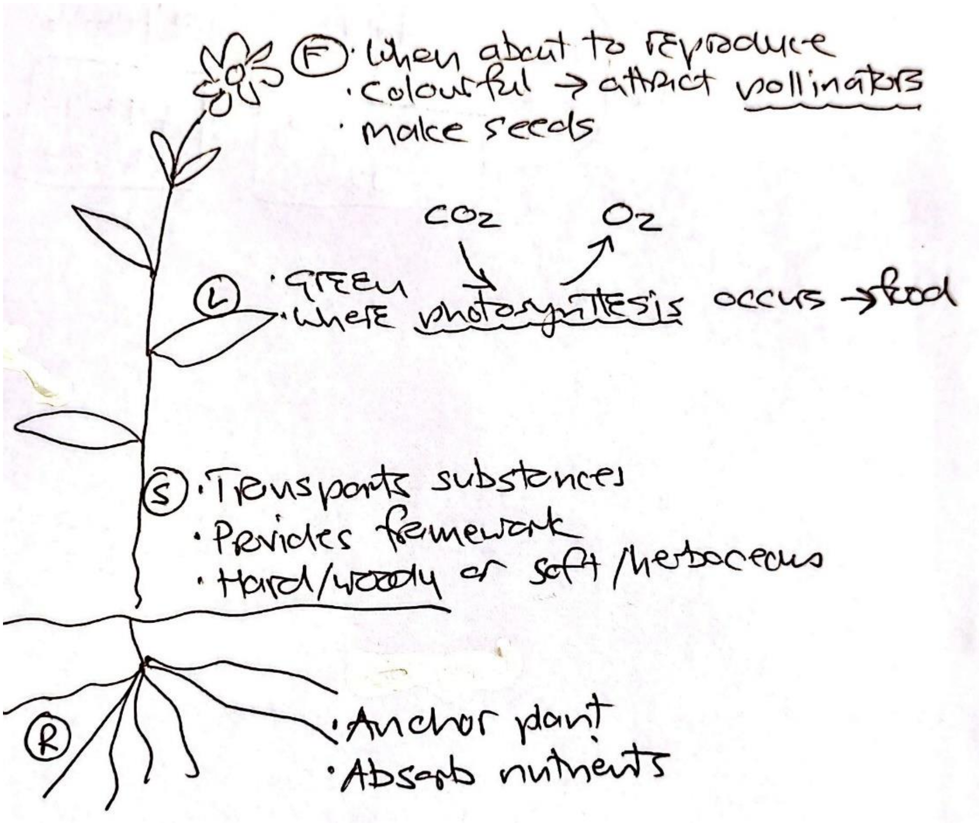
P59

2.5.2 This is a deliberately complex process with many things going on. There are **things** and their states, there are **decisions** – "if this then …"; and there are **processes** - things that happen. Bearing in mind the importance of significance and our built-in drive to find patterns I've decided to put the states in blocks, decision with a question mark, and the processes squiggly underlined. I'm mindful that a process is occurring over time (suggesting left to right flow) but things are also moving up and down, so I've decided to reflect that in the flow diagram by the relevant arrows suggesting the physical up-down. I've also noticed that three processes occur at one point. I've decided that this suggests a list, so I've numbered these.



P61

2.6.1 In this I've drawn a hypothetical plant with just enough detail to show the main parts and low attention to detail. I've chosen to list the annotations again. I've also tried to use shorter language and my own words, e.g., "<u>anchor</u> the plant". I've decided to squiggly underline subject jargon/term that I might want to check up on. Photosynthesis was presented as a process, hence I've made use of flow diagram thinking. Food out, along with CO_2 in and O_2 out. I could have added colour to this but I colour of plant parts is such common knowledge, so I didn't waste time on it.



P65

2.7.1

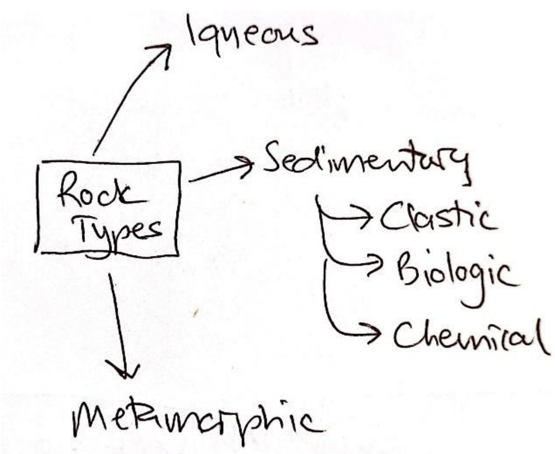
- The three main types of rock
  A= OK (3)   B= huh? There were 6 bullets. Guesses. (1 maybe? Not 3 though)
- Three kinds of sedimentary rock
  A = Sure – a, b, c (3)   B = No idea (0?)
- What do these have in common?
  A = Has to think, but "they are all sedimentary" (2)   B = no idea (0)

Student A won't think there was anything wrong with the test. The questions were a mix on content though the last one they had to use the doc hierarchy to think a bit. Likely to feel the preparation was worthwhile.

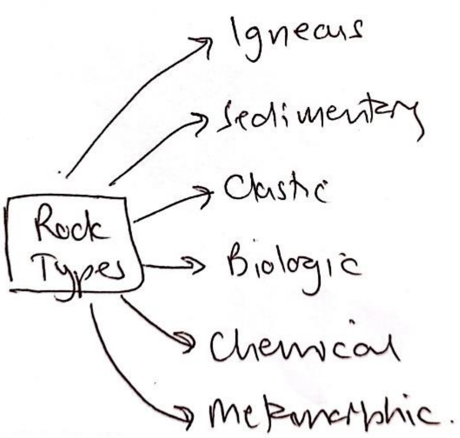
Student B will notch it up as another failure reinforcing their stupidity or inability to study, even though they may have studied hard. Or Student B may blame the teacher for not having taught them the content or asking questions that weren't in the notes. Might feel despondent. If they don't interrogate their result with their teacher, they will likely never know that their concept of the doc hierarchy and their summarising of it is the problem.

2.7.3 The diagram on the left is reminiscent of a Mind Map framework in that it is a visuospatial layout ready to be populated with content. More about Mind Maps later. The diagram could be rendered as just the headings given using the same indentation/numbering if a sequential summary is to hang on it. Note in both cases free hand; no rulers.

Student A



Student B



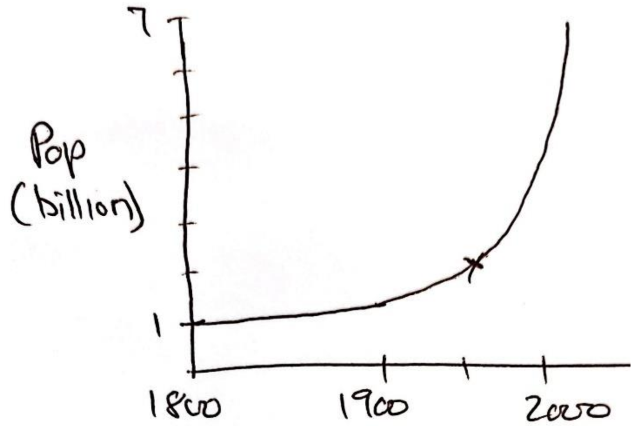
P67

2.8.1 Below is the table. The bulk of it is about what the rocks are made from, but I've decided to split out the Uses descriptor. I've also decided to reinforce that the uses are for biologic only by including an arrow. I've given the sedimentary block a visual layout that makes the subtypes stand out by using indenting. I could have highlighted the 5 examples, just as I underlined the distinction between fossil fuels trapped in sed rocks vs those that are sed rocks. All these decisions increased my level of processing of this content. I had to be engaged to do this. Again, no ruler required.

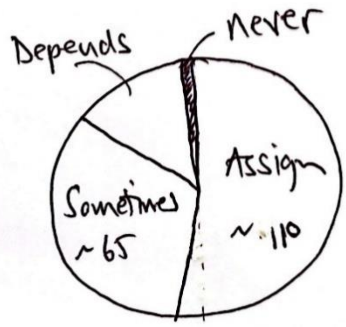
	Igneous	Sedimentary	Metamorphic
Made from?	Melted rock. Inside eg granite Surface e.g. basalt	Clastic: broken rock e.g. sandstone Biologic: Dead things Chemical: Dissolved chems e.g. stalagmite stalactites Cango Caves	Rocks changed by heat/pressure
Uses		Biologic: Fossil fuels. Coal <u>is</u>; oil/nat gas <u>trapped</u>	

P69

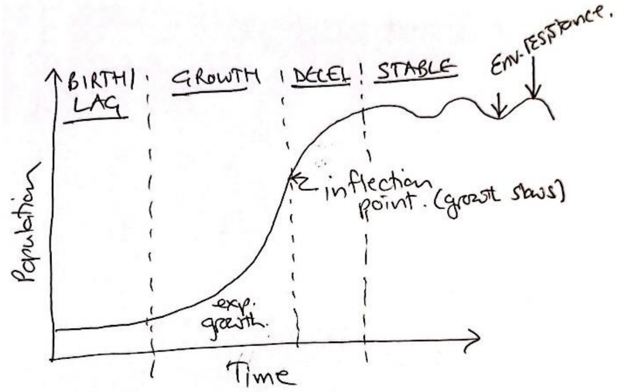
2.9.1 In this graph I've had to think about the scale (a skill in itself) but I can now see visually what the data look like.   Note that I've only indicated key values (1 and 7 on the pop axis; omitted 1950 because I made it halfway between 1900 and 2000)



2.9.2 Here I've had to imagine what the segments look like. The 'third' for 'sometimes' is a third of the *whole* circle. While I was doing this mentally calculated what the actual figures as ratios from 200. 'Depends' I can now see is around 1/6th. This is visual but the processing has meant I've also used a whole lot of maths skills. If I struggle with these graphs it's a good flag that I might need some help.



2.9.3

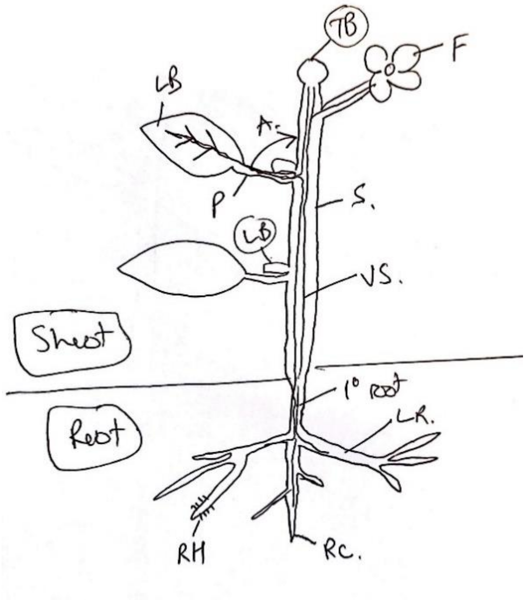


P71

2.10.1 The colours aren’t accurate. The thickness of the layers in huge exaggerated so we can see them. The eyelids are stylised. The capillary network is far more complicated in reality. The blood capillaries aren’t really red and blue; this is simply a colour convention. The conjunctiva is missing. ‘Vitreous’ is not a complete label.
The drawing could have been done from a different angle. The labels don’t have be neatly left and right. The label lines could have gone at an angle.

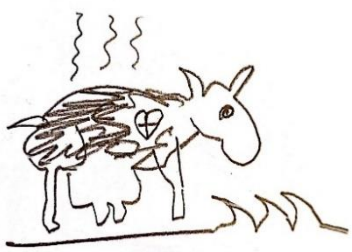
P73

2.10.2 To this diagram, I first studied the original and counted the labels. I then drew the diagram without looking at the picture and made sure I had 12 labels. I decided to use label abbreviations because it makes me think more, embedding retrieval practice. Full labels don’t have this benefit. I had two ‘LB’s so I circled both buds – LB and TB. The vascular system I could have done in a separate colour because all the lines can get confusing. Note that leaf axil and axillary bud both pointed to the same thing. I had to clarify that to ensure I had the correct understanding, and my diagram better reflects the difference.



P75

2.10.3 Here is my fantastic drawing of a cow-inspired mammal! You can see all the features in the list. Importantly, it is based on the first thing that came to *my* mind, and it took all of 90 seconds to draw. Time pressure keeps focus and leverages our innate associations.



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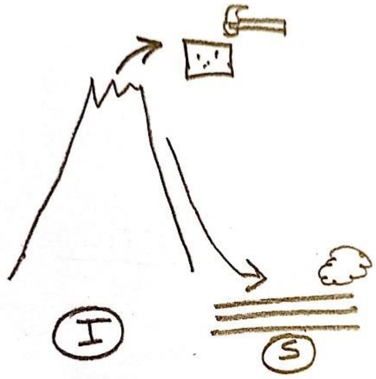
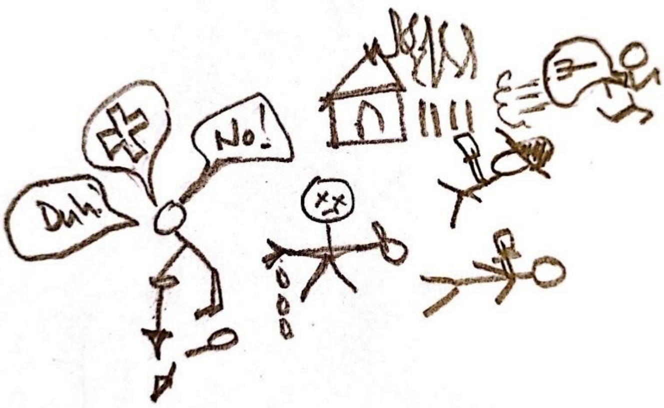
2.10.4 This picture requires some explanation. The volcano is obvious. But I drew a rigid square to represent uniformity and the dots inside are crystals. The hammer means hard. In sedimentary rocks the arrow shows erosion down a mountainside forming layers in which there are no inclusions. To contrast with the hammer, I drew a ball of cotton wool for 'soft'. Finally note that this is a single picture despite the two rock types. Combining strengthens association between the concepts more than two separate diagrams would and in this case it was easy. While the picture may not be self-explanatory to outsiders, they are to me and for my purposes, and that's the only person that matters in a test!



P77

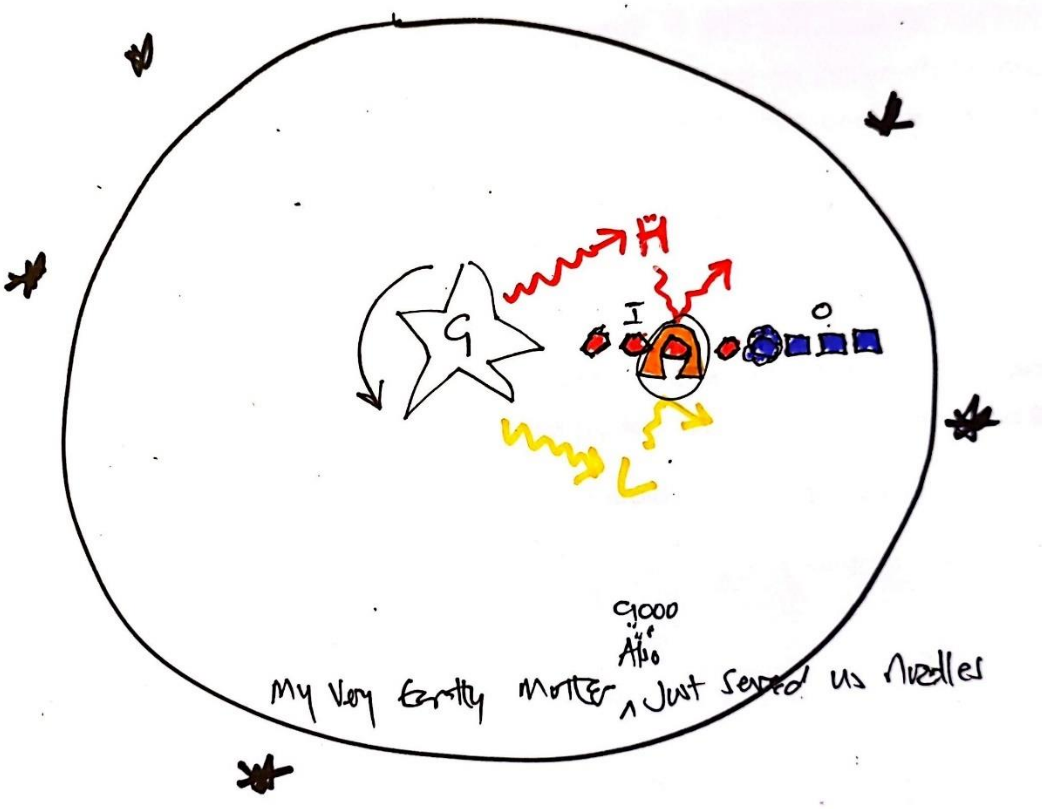
2.10.5 Actually this turned out to be quite easy. Below is a single scene of chaos (chunked). Remember that the order of things doesn't matter. Again the drawing is done deliberately fast to maintain focus and prevent distraction on the unimportant.
Explanation:

- Left hand figure: speech bubbles indicate disobedience ("No!"), feigning illness (the medical cross), and being cheeky (Duh!). The person is leaning on a spade (lazy) and there is a broken spade too (broke stuff).
- Next figure right: x-eyes and holding a bottle (got drunk) and bleeding from hand (hurt themselves).
- Lying figure bottom: killed by own hand holding a knife (suicide)
- Top lying figure. Distingished by a hat indicating the master. Also dead by a knife (murdered masters).
- Top: House and field of crops on fire (set fire to them)
- Top right running figure (ran away) holding a bag with a fork (stole kitchen utensils)



2.10.6 For a change I used colour to show how the colour is representative of something. Blue=cold, red= hot, yellow = sun's rays.

Solar system (circle) with stars outside and star (sun) at the centre. Sun held together by gravity ('G) and rotates (circular arrow). Gives off rays (wavy lines) of heat ('H'; red) and light ('L'; yellow). Four planets closest to sun are inner planets ('I'), hot (red) and rocky (angular shapes). Outer ('O') four are cold (blue). First is gaseous (airy shape) vis icy (ice-block-shaped). Earth in goldilocks zone (blonde hair added). Layer around the earth (atmosphere) maintains just right conditions by reflecting light and heat (deflected arrows). Planets order given by the acrostic embedded in the diagram. Acrostic modified by 'Also' (asteroid belt) dots (asteroids) and Ceres the biggest at 100km across (C1000)



P83

1. 1.1 F (can be done with no understanding) 1.2 F (pages of writing out have low value) 1.3 T (you can't transform if you aren't engaged) 1.4 T (probably, unless CERI means something to you) 1.5 F (pretty never. Colour use must be judicious) 1.6 T (probably the most powerful way if we had to imagine the diagram for ourselves)
2. 2.1 annotate 2.2 patterns 2.3 hierarchy 2.4 flow diagrams 2.5 Graphs 2.6 Tables 2.7 annotation 2.8 chunked 2.9 fast 2.10 annotation

Challenge: A FLAG PATCH  >>>

Acronym | Flow diags | Lists | Acrostics | Graphs | Pictures | Annotation | Tables | Colour | Hierarchies

P87

2.1 There is lots you can say. Trigger Maps build in all of what we have been talking about. Let's consider from the very beginning
  - They take effort but are achievable
  - Doing them and being able to remember is fulfilling and adds confidence
  - They lend themselves to goal setting and APs if you do them progressively. Its easy to set SMARTA goals and APs involving trigger maps.
  - It's up to you how you do them; their development is in your control and reflects *your* thinking
  - They re quite creative so hopefully you feel at least a bit like doing them
  - They can be part of good habit formation
  - You can do them daily
  - They force retrieval because the actual content is not in front of you – only triggers that remind you of what you know
  - Triggers maps are pictures (we all think in pics), they are organised (logical storage) and the triggers rouse our curiosity ("What did that stand for again?"). The diagrams are rich is elements that are all significant, they can stimulate interest when you finally start getting to grips with the content, and they leverage repetition as you revise from them.
  - They are high on association; you are always linking things
  - They for a big picture and you are always seeing the whole even if you are focussing on a part
  - They require understanding and demonstrate it to you
  - Triggers can consist of all the tools we learnt about and the entire map is one huge transformation exercise
  - They represent time well spent studying. You never redo them so you're not wasting time in the long run.
  - Etc…

2.2 These are valid concerns and only practice will overcome them. Also, working with a teacher as part of your SMARTA accountability and sharing your challenges.
  - **They look hard to do**. Challenging yes, but that's a good thing. At first, surfacing your way of thinking – your natural associations – may take longer.
  - **I'm not an artist**. You can see by my diagrams that you don't have to be! It's the thinking that your drawing triggers that matters.
  - **They'll take long**. Studying always takes time. Think of how long you'll spend writing out or memorising, but how much more effective and less laborious this will be.
  - **It's risky to change**. It is. Start small. Build your confidence by trying it one section of work or in a low-stakes situation.
  - **What I do works for me**. What are the underlying principles in your method that make it work so well compared to the principles for trigger maps? Bear in mind that for many students what 'works' at school fall horribly short at university.

P89

2.3 A great attempt that illustrates many of the principles. Good:

- Layout: Clear layout. Full but not cluttered or confusing. Radiating layout is appropriate. Doc hierarchy is reflected in the heading styles
- Colour: Used judiciously
- Evidence of **graph** which is **annotated** (product life cycle), **acronyms** (PATPGS/WCTPGS), a **table** (SWOT), a trigger **picture** (factors affecting business), a **list** (Strategy), **flow diagram** element (Place)

Improve:

- 'Choosing a suitable product' is not chunked and uses standalone pics which is not good practice.

P93ff

Regarding a study planner for exams, see the separate Excel file on this site for you to download and complete. You can modify it if you wish and know how.